

1.9.3

EE25BTECH11014 - Bhoomika Lokesh

Question: $AOBC$ is a rectangle whose three vertices are $(0, -3)$ $(0, 0)$ $(4, 0)$. The length of its diagonal is_____.

Solution: Given the points **A**, **O** and **B** :

Point	vector
Point A	$\begin{pmatrix} 4 \\ 0 \end{pmatrix}$
Point O	$\begin{pmatrix} 0 \\ 0 \end{pmatrix}$
Point B	$\begin{pmatrix} 0 \\ -3 \end{pmatrix}$

TABLE 0: Vectors of the points.

Determining the Coordinates of Point C:

$$\vec{OA} = \begin{pmatrix} 4 \\ 0 \end{pmatrix} \quad \vec{OB} = \begin{pmatrix} 0 \\ -3 \end{pmatrix} \quad (0.1)$$

Since **C** is opposite to **O** in the rectangle,

$$\vec{OC} = \vec{OA} + \vec{OB} = \begin{pmatrix} 4 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ -3 \end{pmatrix} = \begin{pmatrix} 4 \\ -3 \end{pmatrix} \quad (0.2)$$

$$\therefore \mathbf{C} = \begin{pmatrix} 4 \\ -3 \end{pmatrix} \quad (0.3)$$

We know that the length of the diagonal vector is magnitude of the vector $\vec{OC}$.
Let $\mathbf{P} = \vec{OC}$

$$\mathbf{P} = \begin{pmatrix} 4 \\ -3 \end{pmatrix} \quad (0.4)$$

$$|\mathbf{P}| = \sqrt{\mathbf{P}^T \cdot \mathbf{P}} \quad (0.5)$$

$$\mathbf{P}^T = (4 \quad -3) \quad (0.6)$$

$$\mathbf{P}^T \cdot \mathbf{P} = (4 \quad -3) \begin{pmatrix} 4 \\ -3 \end{pmatrix} = 4^2 + (-3)^2 = 16 + 9 = 25 \quad (0.7)$$

$$|\mathbf{P}| = \sqrt{25} = 5$$

Therefore the lenght of the diagonal is 5.

See Fig.1,

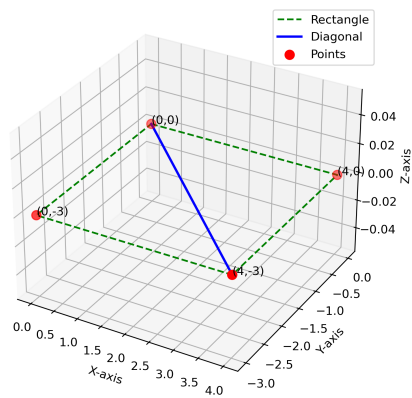


Fig. 0.1: Graph